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Course Code: ITELEC1

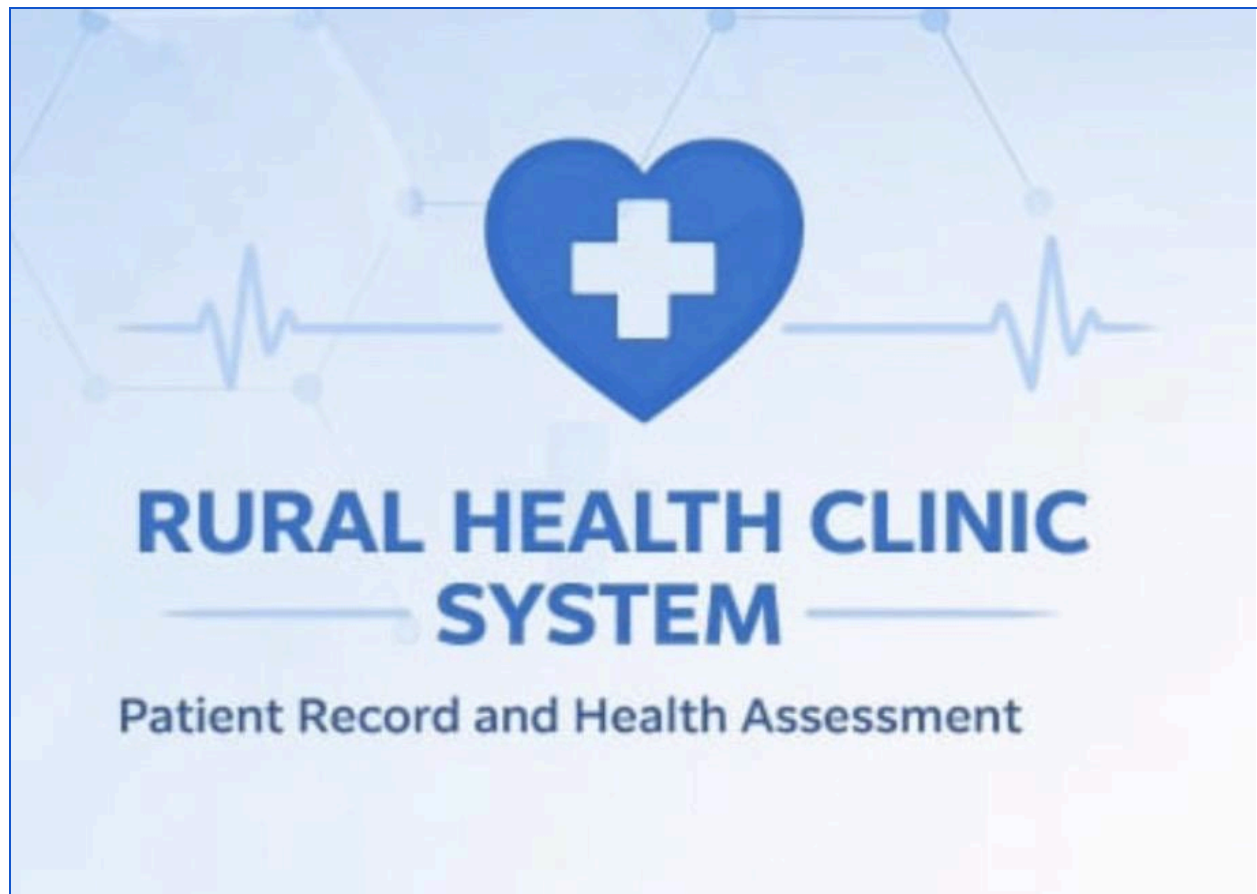
Course Name: IT Elective 1 (.NET/C#)

Program/Year Level: 2.7BSIT

Semester/Term: 2nd Semester, Cycle 2

DOCUMENTATION

SDG 3: RURAL HEALTH CLINIC SYSTEM



Introduction

Being healthy is the key to improving our well-being, especially in rural areas where access to health services is limited. Many clinics in rural areas are still operating in a traditional way such as pen and paper for recording several patients, which results in losing track of the files, slow operation and low quality of services.

To address these problems, we developed the Rural Health Clinic Patient Record System, a VB.NET Windows Forms Application, to enhance the management of patient information in rural health care settings. In addition, the system is designed to provide a secure, organized, and efficient way of storing, retrieving, updating, and generating patient records using ADO.NET database connectivity and reporting tools. This system can assist healthcare staff to reduce their manual workload, make it accurate, and deliver fast healthcare services by digitizing patient records.

This project aligns with United Nations Sustainable Development Goal (SDG) 3: Good Health and Well-Being. The project supports the goal of helping healthcare providers easily access and strengthen healthcare systems in underserved rural communities.

Requirement Analysis

Functional Requirements (FR)

FR1 - Data Persistence

The system allows users to add, view, update, and delete patient and consultation records using an SQL Server database.

FR2 - Business Logic

The system can calculate BMI and display the patient's health status.

FR3 - Reporting

The system can generate patient and consultation reports and export them as PDF using QuestPDF. (ReportViewer wasn't used, it's not compatible with my PC)

FR4 - Security

The system has a secure login system using your username and password credentials to authenticate.

Non-Functional Requirements (NFR)

NFR1 – UI Consistency

The system uses a clean blue theme and a user-friendly interface with proper naming conventions.

NFR2 – Maintainability

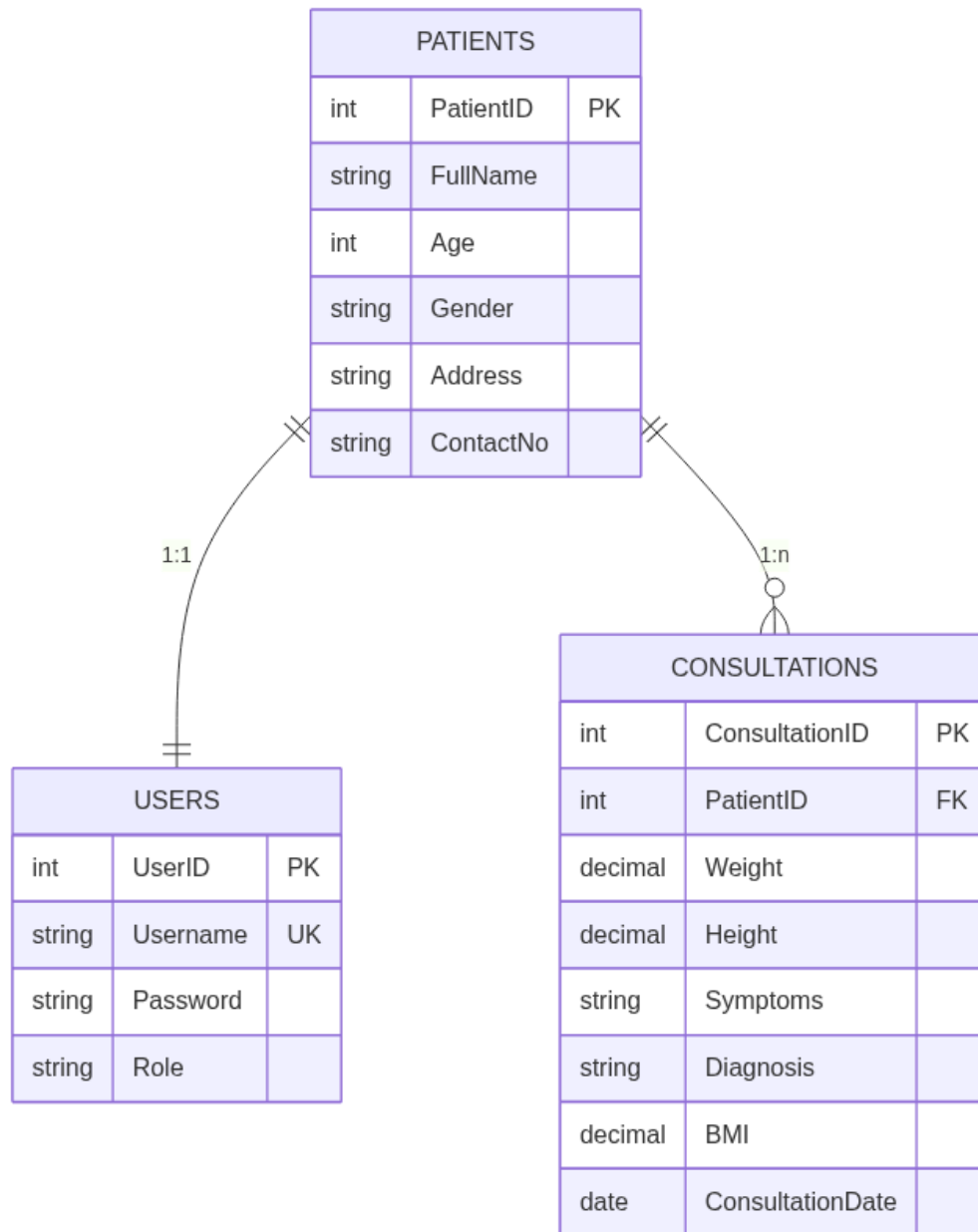
The code is organized and documented.

NFR3 – Reliability

The system uses a dedicated connection class VB and error handling to ensure a stable connection between forms.

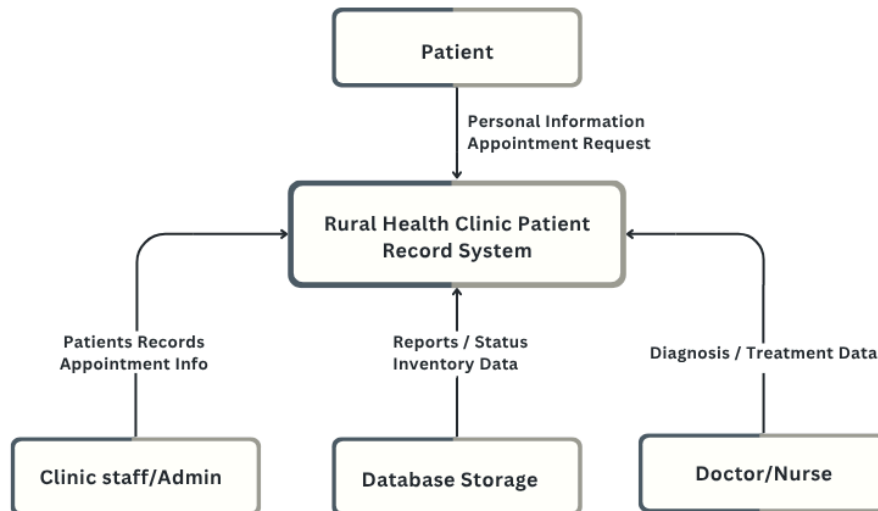
Database Design

- Entity Relationship Diagram (ERD)



System Architecture

DFD LEVEL 0

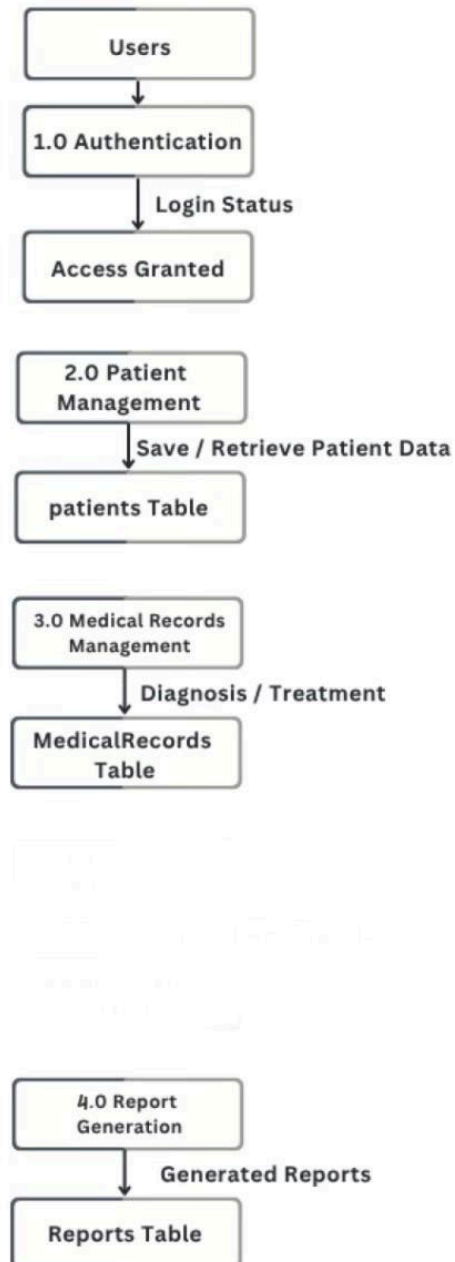


Data Flow Diagram Level 0 of the SDG 3: Rural Health Clinic System

DFD Level 0 of Rural Health Clinic Patient Record System. The DFD Level 0 shows the high-level flow of the system. Like Patients, the system also involves the clinic staff and healthcare providers in providing and getting patient information, consultation records, and reports. The system processes and stores data to make the clinic works easier.

System Architecture

DFD LEVEL 1



Data Flow Diagram Level 1 of the SDG 3: Rural Health Clinic System

DFD Level 1 of Rural Health Clinic Patient Record System DFD Level 1 depicts the major processes of the system, which include user authentication, patient management, recording of consultation, and report generation. The system communicates with the database for data storage and access, enabling efficient handling of patient information and health records by the clinic staff.

Individual Contribution

CONTRIBUTION	MEMBER
Form Design	Claros
Visual Studio Execution	Chen
SQL	Chen
Documentation	Both
Presentation	Both